PAVITHRAN G

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PROFESSIONAL SUMMARY

Artificial Intelligence and Machine Learning undergraduate undergraduate specializing in Agentic AI, hybrid RAG systems, and full-stack development. Skilled in building production-ready AI solutions and scalable backend systems, with strong DSA fundamentals and 200+ problems solved.

EDUCATION

K.S. Rangasamy College of Technology **2023 – 2027**
B.E. in Artificial Intelligence and Machine Learning (CGPA: 8.95 / 10.0) *Tiruchengode, India*
Sengunthar Matriculation Higher Secondary School **2023**
Higher Secondary Certificate (HSC) *Tharamangalam, India*

TECHNICAL SKILLS

Programming Languages: Python, Java, SQL
AI & Machine Learning: Machine Learning, Agentic AI, Generative AI, LangGraph, LangChain, RAG
Backend & APIs: FastAPI, REST API Design
Databases & Automation: PostgreSQL, MySQL, n8n, UiPath
Concepts: Data Structures and Algorithms, Agile Methodology, System Design
Developer Tools: Git, GitHub, Linux, Google Colab, VS Code

EXPERIENCE

Software Developer **Feb 2026 – Apr 2026**
Mecandria IT Services and Solutions *Intern*

- Engineered a full-stack healthcare management application, contributing across the SDLC from requirement analysis and REST API development to frontend integration, database design, testing, and production deployment.
- Deployed and configured cloud-hosted applications on DigitalOcean Ubuntu servers, integrating FastAPI, PostgreSQL, Nginx, and HTTPS with a focus on secure configuration and reliable service delivery.
- Troubleshot production issues across API connectivity, database permissions, server configurations, and SMTP email workflows, strengthening application reliability and deployment stability.

PROJECTS

Lexicon AI – Agentic Hybrid RAG Knowledge Engine | *Python, FastAPI, LangGraph, ChromaDB, BM25*

- Architected an Agentic Retrieval-Augmented Generation (RAG) system with a self-correcting **LangGraph CRAG loop** to rewrite queries and re-retrieve context on low confidence, preventing hallucinations.
- Implemented **Cascade Hybrid Retrieval** (BM25 sparse + ChromaDB BGE dense + Cross-Encoder reranking), boosting retrieval relevance by 30–40%.
- Orchestrated multi-provider Large Language Model (LLM) failover (**Groq, NVIDIA NIM, OpenAI, Ollama**) integrated with sub-100ms FastAPI SSE streaming.

Motion Capture System using MediaPipe and Unity | *Python, OpenCV, MediaPipe, Unity, C#*

- Developed a real-time human pose recognition pipeline utilizing OpenCV and MediaPipe for spatial keypoint tracking.
- Built a low-latency socket bridge transmitting 3D coordinate vectors into Unity for live skeletal binding.
- Implemented coordinate smoothing filters to eliminate joint jitter and enhance interactive 3D visualization.

CERTIFICATIONS

- **Microsoft Certified: MLOps Engineer Associate** (2026)
- **Redis Associate Developer**(2026)
- **Oracle Certified AI Agentic Foundation Associate** (2026)
- **Redis Associate Cloud Operator**(2026)
- **Claude Certified Architect** (2026)
- **Oracle Certified AI Foundation Associate** (2025)

ACHIEVEMENTS

- **TCS CodeVita Season 13**: Achieved Global Rank **1491** among 100,000+ international participants.
- **TN-Skills 2025**: Qualified Level-1 and Level-2 in Software Application Development.
- **Competitive Programming**: Solved **200+ Data Structures and Algorithms (DSA) problems** across LeetCode; earned **5-Star Badges** in Java & Python on HackerRank.